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Empirical Research Paper

Perceiver Positivity Differences Originate in Early-Stage Impression Formation: Evidence From a Recognition Task

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Abstract

People differ in how positively they perceive others at first glance. The present research examined whether such positivity differences are already evident during early stages of impression formation, under conditions that do not require explicit evaluative judgment. We developed a recognition-based paradigm in which participants read vignettes describing unfamiliar targets and later completed an unexpected recognition task. In Study 1 ($n = 312$), participants differed in the valence of traits they recognized; those who recognized more positive traits reported greater liking for the targets and scored higher in agreeableness, conscientiousness, and age. Study 2 ($n = 837$) replicated these findings, introduced a "don't know" response option to reduce guessing, and demonstrated moderate overlap with rating-based measures of perceiver positivity. Together, the results indicate that perceiver positivity differences can emerge during early impression formation and reflect stable individual differences.

Keywords

perceiver effects, person perception, spontaneous trait inference, impression formation, positivity

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Introduction

Individuals systematically differ in the way they judge other people. These individual judgment styles are known as *perceiver effects* within the Social Relations Model (Kenny, 2019), and the most important driver of perceiver effects is the global positivity with which individuals typically view others (Heynicke et al., 2021; Srivastava et al., 2010; Wood et al., 2010). For instance, a positive perceiver might have the tendency to view others as highly friendly, smart, trustworthy, and attractive, whereas a negative perceiver might typically judge others lower on these attributes. Individual differences in perceiver positivity have been found in different social contexts (e.g., with strangers or roommates) and across judgment settings (e.g., with and without face-to-face interactions; Rau, Carlson, et al., 2021). Moreover, positivity differences are temporally and cross-situationally stable (Rau, Lawless DesJardins, et al., 2022), suggesting they are an important individual difference variable.

One question that has remained open in research on perceiver effects concerns the stage of impression formation at which positivity differences emerge. Specifically, do such differences primarily reflect later, intentional judgment tendencies (e.g., choosing to judge others benevolently or critically), or are they already evident during earlier stages of information processing as perceivers encode personality-relevant cues? Importantly, these processes are not mutually exclusive and may jointly shape impressions in everyday social perception. The present research addresses this issue by moving beyond the common practice of computing perceiver effects from explicit Likert-type ratings and instead examining positivity differences using a newly developed recognition-based paradigm. The rationale is that if systematic positivity differences arise during early stages of impression formation, they should be detectable even under conditions that do not require explicit evaluative judgments, and thus leave traces in memory. To test this idea, we designed a recognition task that operationalizes perceiver positivity in terms of the average valence of trait information

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that perceivers recognize after being exposed to balanced personality descriptions of unfamiliar targets.

Measuring Perceiver Positivity

The common method to measure perceiver positivity is to explicitly ask participants to provide ratings about several targets on several personality traits (e.g., "This person seems dependable, self-disciplined"; "This person seems calm, emotionally stable") on a scale ranging from 1 (*does not apply at all*) to 8 (*applies perfectly*; Rau, Nestler, et al., 2021). Individual positivity scores are then computed by aggregating the ratings each perceiver has provided across all targets and traits (after reversing ratings for undesirable traits). Based on this operationalization, researchers routinely interpret differences in the perceivers' judgments as indicators of differences in the perceivers' perceptions of others.

However, perceivers may have judgment tendencies that are independent of their actual spontaneous impressions of targets. For instance, someone with a self-concept of being highly agreeable might adhere to a norm of judging others benevolently, regardless of actual impression. Similarly, female gender roles might foster more communal, selfless, and other-concerned behaviors—including positively biased judgments (Eagly, 1987; Winquist et al., 1998). In such cases, the spontaneous impression might be unaffected by traits like agreeableness or gender; rather, the key differences may reside in deliberate evaluative judgment.

Because rating-based approaches cannot disentangle early impression processes from later, explicitly evaluative judgment tendencies, we sought alternative methods. A well-established technique from experimental psychology is the false recognition paradigm (e.g., Rim et al., 2013; Todorov & Uleman, 2002, 2003). In this paradigm, participants first read behavioral sentences describing various targets (e.g., "Bea cleaned up her apartment all afternoon"). Later, they complete a recognition task, judging whether certain trait adjectives (e.g., "neat") had been presented previously. Research using this method has shown that people spontaneously infer traits even when not explicitly instructed to do so (Winter & Uleman, 1984), leading to false recognition of traits implied—but not stated—in the descriptions. We adapted this approach to examine whether perceivers differ in the valence of the traits they spontaneously encode.

The Present Research

Across two studies, we developed a recognition-based paradigm that captures the valence of trait content perceivers (correctly or erroneously) remember after being exposed to balanced personality descriptions of unfamiliar targets. We reasoned that if perceiver positivity is rooted in early stages of impression formation, these tendencies should manifest even in the absence of any explicit evaluation prompt. Importantly, the paradigm does not eliminate intentional cognition altogether. Because encoding is self-paced and unconstrained, participants may engage in varying degrees of reflective processing. However, the paradigm crucially differs from rating-based approaches in that it does not require participants to form or express an explicit evaluative judgment. Thus, it primarily constrains evaluative demands at the judgment stage rather than fully controlling encoding processes. Study 1 served as a proof of concept, establishing that perceivers systematically differ in the positivity of trait information they recognize and that this spontaneous positivity correlates with relevant validation variables. Study 2 refined the recognition task psychometrically and better distinguished spontaneous encoding from strategic guessing or intentional recall. Further, Study 2 used additional validation measures and recruited a more demographically diverse sample to ensure generalizability.

All procedures were approved by the ethics committee of the University of Münster (protocol number 2021-25-RR). Neither study was preregistered. We report how we determined the sample size, all data exclusions, all procedures, and all measures in both studies. Further, we provide raw data, analysis scripts, study materials, pretest results, and additional analyses online (https://osf.io/64389/).

Study 1

We designed Study 1 to establish that perceivers vary in terms of the positivity of information they tend to encode about strangers at all. In a second step, we sought to explore the psychological meaning of these positivity differences. To this end, we examined how positivity was associated with recognition accuracy and with variables that have previously been used to qualify the nomological network of positivity in perceiver effects research.

One such variable is how much a perceiver explicitly says they *like* the targets. Arguably, "likers" might preferentially process positive personality information about others, whereas "dislikers" preferentially process negative information. In perceiver effects research, positivity and liking have been shown to be strongly associated (Heynicke et al., 2021). As another set of validation variables, we considered various personality dimensions, specifically, the Big Five (Goldberg, 1990) and narcissistic admiration and rivalry (Back et al., 2013). Among these dimensions, agreeableness and narcissistic rivalry have consistently been linked to positivity in perceiver effects research such that highly agreeable and non-rivalistic perceivers tend to rate others more positively (Rau, Carlson, et al., 2022; Rau, Nestler, et al., 2021; Wood et al., 2010). Finally, as a third set of validation measures, we considered sociodemographic variables (i.e., age, gender, and education). In previous research, female participants have repeatedly been found to judge others more positively than male participants (Rau, Nestler, et al., 2021; Srivastava et al., 2010; Winquist et al., 1998; Wood et al., 2010) and a

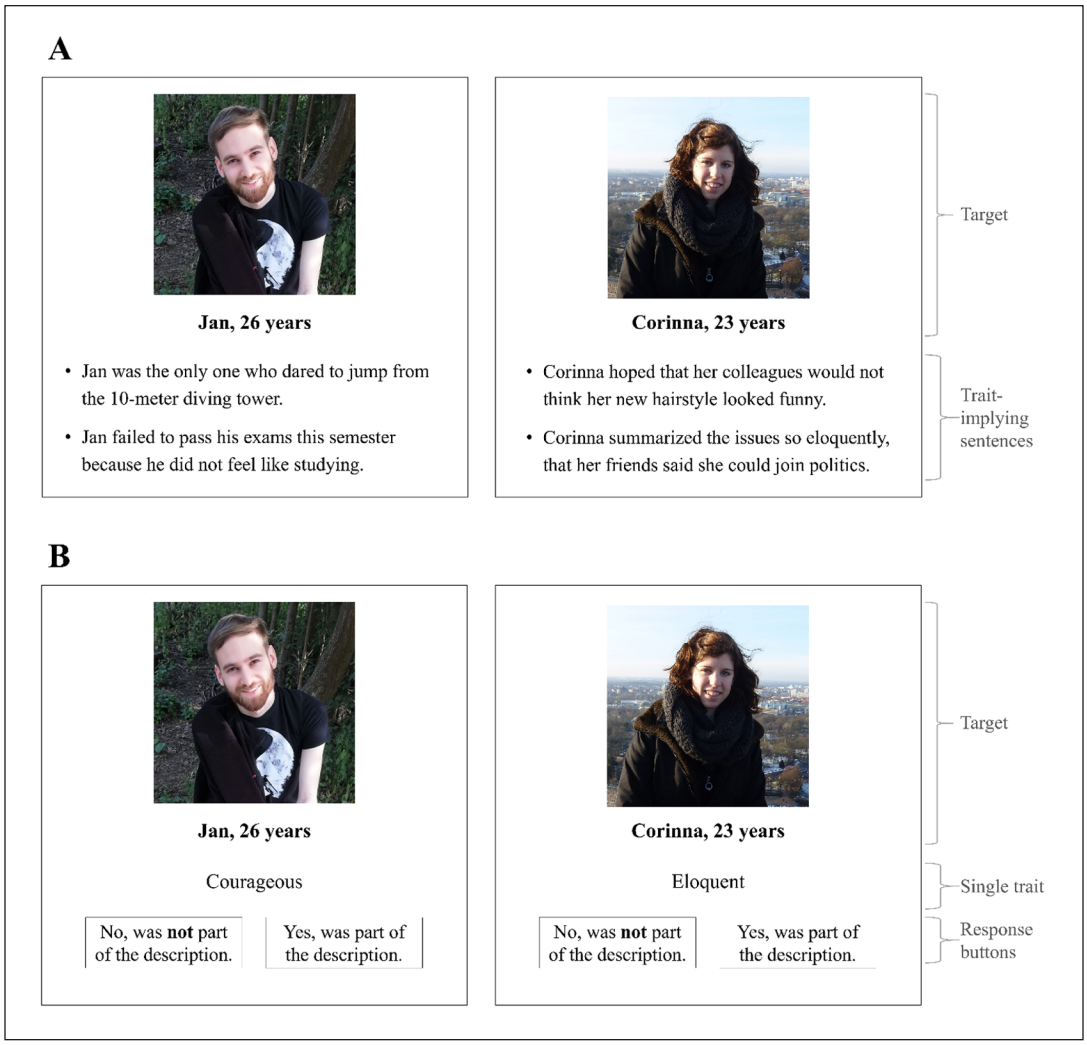

Figure 1. Example vignettes of the recognition task. In the study phase, each target was presented with two trait-implying sentences (panel A). In the test phase, each target was presented with a single trait (panel B). The trait-implying sentences and traits are adaptations of the original German study material (see OS 1). For Jan, the traits *courageous* and *lazy* are implied. For Corinna, *insecure* is implied, and *eloquent* is explicitly presented. The targets displayed above are used for illustration but were not part of the original task.

higher level of education was associated with more negative perceptions of others in one previous study (Rau, Nestler, et al., 2021). Study 1 was not preregistered.

Methods

Recognition Task

Stimuli. To create the stimuli for the recognition task, we generated 18 targets (half male, half female) as presented in Figure 1. For each target, we selected a name, an estimated age, and a photo in the typical style of social media profile pictures (retrieved from the license-free image website http://freeimages.com). To mirror the current composition of the German population in the target pool (Bundeszentrale für Politische Bildung, 2020), four of the targets were chosen so as to represent ethnic minorities (two Arabic, one Turkish, one Slavic). We next selected adjectives that would form the basis for the trait-implying sentences shown in the vignettes and that would be presented to participants during the test phase. We aimed at covering both desirable and undesirable traits across different personality domains. Therefore, we selected 36 German trait adjectives that were approximately equally distributed across the Big Five domains according to Ostendorf (1990). Specifically, we selected eight traits for extraversion and seven traits for agreeableness, conscientiousness, emotional stability, and openness to experience, respectively, with an equal number of desirable and undesirable traits across the five factors.

Finally, following established paradigms from the study of spontaneous trait inferences (Uleman, 1988), we generated behavioral sentences for all 36 traits. Twelve sentences explicitly contained the relevant traits as adjectives or

adverbs (e.g., "Corinna summarized the issues so *eloquently*, that her friends said she could join politics"). In contrast, the remaining 24 sentences implied the relevant traits rather than stating them explicitly (e.g., "Jan failed to pass his exams this semester because he did not feel like studying" implies *lazy*). The number of sentences for desirable vs. undesirable traits was balanced. In a pretest, 33 participants (10 male, 13 female; $M_{age} = 25.7$, $SD_{age} = 5.6$) read the 24 trait-implying sentences and came up with up to three synonymous trait adjectives describing the target person in an open-response format. We found considerable consensuses for all 24 traits ($M = 66.79\%$, $SD = 19.44\%$), which corroborated that the behavioral sentences indeed implied the relevant traits. The original data and R script of the pretest can be retrieved from the OSF.

Study Phase. In the study phase, participants were presented with the 18 vignettes in randomized order (see Figure 1A for examples). As a cover task, they were requested to speculate which music genre they believed the targets to prefer by selecting 1 out of 10 response options. In each vignette, the target was described by one desirable and one undesirable sentence, thus providing personality information that was overall neutral. Half of the vignettes began with the desirable sentence, the other half with the undesirable sentence. In each vignette, the two sentences covered two different Big Five dimensions (e.g., *courageous* as an indicator of high extraversion and *lazy* as an indicator of low conscientiousness in the example vignette of Jan in Figure 1A).

Test Phase. In the unannounced test phase, participants were asked to decide whether or not a given trait had been part of the target's description (Figure 1B). For each target, four different traits were presented, resulting in a total of 72 trials, each of which was shown on a separate survey screen. The four traits featured the two traits from the corresponding description (e.g., *courageous* and *lazy* for Jan) and two traits from other targets' descriptions (e.g., *eloquent* and *nervous* for Jan, where *eloquent* was actually part of Corinna's description, and *nervous* was actually part of another target's description). The four traits per target each covered distinct Big Five domains[1] and appeared successively in a randomized order before moving on to the next randomly selected target. In 24 of the 72 test trials, targets were systematically paired with traits that had actually been implied in their descriptions during the study phase (correctly paired implied trials; *courageous* and *lazy* for Jan; *insecure* for Corinna). In another set of 24 trials, targets were randomly and incorrectly paired with traits that had been implied for a different target (falsely paired implied trials; e.g., *courageous* for Corinna; *insecure* for Jan). In the remaining 24 trials, targets were paired with traits that had actually been presented (explicit trials). Of these, 12 trials were correctly paired (e.g., *eloquent* for Corinna) and 12 were incorrectly paired (e.g., *eloquent* for Jan).

Participants. Participants were recruited via digital bulletin boards and mailing lists of the University of Münster, and via posts on social media. To be included in the study, participants had to be at least 18 years old, have very good knowledge of the German language, and complete the entire survey. They were compensated by a lottery of 10 times 50€. Psychology students could additionally earn course credit. We planned to collect data from at least 200 individuals because this sample size yields 80% power to detect moderate associations of $\rho \geq .20$ at an alpha-level of .05 in a two-sided test. Initially, 313 participants completed the study. One participant was excluded because their pattern of responses indicated carelessness (99% "No"—responses in the test phase). The completion of the survey took the remaining 312 participants between 14 and 21 min (interquartile range), with a median duration of 16 min. The final sample ranged from age 18 to age 72 ($M = 32.0$, $SD = 12.0$) with 92 identifying as male (29%), 219 as female (70%), and one person as diverse (< 1%). Most people in the sample were rather highly educated, as is evident from their reports on a 6-point scale (1 = *no graduation* [0%]; 2 = *graduated after 8th grade* [1%]; 3 = *graduated after 10th grade* [4%]; 4 = *graduated from senior high school* [25%]; 5 = *apprenticeship* [21%]; 6 = *bachelor´s degree or higher* [49%]). As is common in Germany, no information about participants' race or ethnicity was assessed. The obtained sample size allowed a statistically powerful first test of the recognition task, with 80% power to detect associations of $\rho \geq .16$ at a two-sided alpha-level of .05.

Procedures. The study was implemented as a self-paced online survey via the platform formr.org (Arslan et al., 2020). Participants were informed that the study investigated the relationship between personality and music preference. After providing informed consent, participants worked on the recognition task as described above. They then completed self-reported personality measures and indicated how much they liked each target. Finally, participants provided sociodemographic information, were debriefed about the actual purpose of the study, and received compensation.

Validation Measures

Explicit Liking. Each target was presented one by one in randomized order. Participants rated "I like this person" on a 6-point Likert-style scale ranging from 1 (*does not apply at all*) to 6 (*applies perfectly*). For each participant, the mean score in liking across targets was computed ($\alpha = .69$).

Big Five. Big Five personality dimensions were assessed via self-reports on the 36 adjectives used in the recognition task. Items read "In general, I am . . ." followed by the respective adjective. They were answered on a 6-point Likert-type scale ranging from 1 (*do not agree at all*) to 6 (*completely agree*). Extraversion was measured with eight items ($\alpha_E = .85$), while agreeableness ($\alpha_A = .62$), conscientiousness

(α_C = .79), emotional stability (α_{ES} = .83), and openness (α_O = .60) were measured with seven items each.

Narcissism. Narcissistic admiration and rivalry were assessed with the Narcissistic Admiration and Rivalry Questionnaire Short Scale (NARQ-S; Back et al., 2013) using a 6-point Likert-type scale from 1 (*does not apply at all)* to 6 (*applies perfectly*; $\alpha_{admiration}$ = .76; $\alpha_{rivalry}$ = .61).

Descriptive statistics and intercorrelations of all validation measures can be found online (OS 2).

Results and Discussion

Preliminary Analyses

To index the valence of each participant's average recognition, we computed a positivity score from the 72 test trials. To this end, we first classified each trial as desirable versus undesirable according to the trait adjective's mean valence rating (retrieved from Britz et al., 2019). Then, we subtracted each participant's number of recognitions (i.e., answering "Yes, was part of the description") for undesirable trials from the number of recognitions for desirable trials. This difference was subsequently divided by the participant's total recognitions to control for individual differences in the general readiness to recognize any trait. For example, if Paula recognized 30 desirable and 15 undesirable traits and Peter recognized 10 desirable and five undesirable traits, both their positivity scores would be 0.33 ([30-15]/45 and [10-5]/15, respectively). Positivity scores can vary between −1 and 1, where 0 indicates a neutral valence in recognitions.[2]

Before turning to our main research questions concerning the positivity score, we sought to confirm that the newly developed task actually tapped the positivity of personality impressions that were formed during the study phase rather than just the positivity of random guesses made during the test phase. For instance, the reason why participants recognized *courageous* as part of Jan's description was supposed to be that they inferred that Jan must be courageous and stored this inference as part of their mental representation of Jan. If this was the case, one would expect higher recognition rates for trials in which adjectives and vignettes were correctly paired (e.g., *Jan* paired with *courageous* and *Corinna* paired with *eloquent*) rather than incorrectly paired (e.g., *Jan* paired with *eloquent* and *Corinna* paired with *courageous*).

As shown in Figure 2, recognition rates were indeed considerably higher for correctly paired as compared to incorrectly paired trials. This was true both for trials in which the given adjective had actually been presented during the study phase (explicit trials, left part of Figure 2) and for trials in which adjectives had only been implied (implicit trials, right part of Figure 2). Note that both of these tests corroborate that the positivity score does not capture the positivity of random guesses. However, only the test regarding explicit trials concerns the question of *accuracy* in a strict sense, because

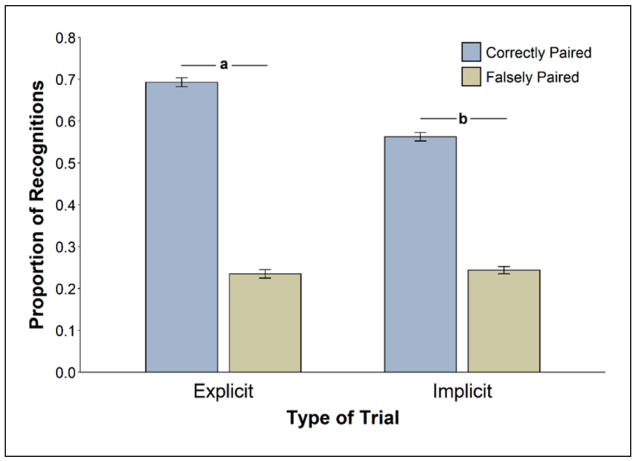

Figure 2. Mean recognition rates (with standard errors) by type of trial in Study 1. Trials with a correct pairing of adjective and vignette were recognized significantly more often than trials with an incorrect pairing.
Note. a: $t(311) = 31.38$, $p < .001$, $d = 2.52$; b: $t(311) = 28.21$, $p < .001$, $d = 1.94$.

responding with "Yes, was part of the description" was technically not correct in any implicit trial (because even if the adjective was paired with the correct target, it was only implied but not actually presented in the description). For this reason, recognition accuracy is commonly computed only for explicit trials, specifically as the proportion of correct recognitions in explicit trials. On average, recognition accuracy was 69%, which falls within the range of accuracies reported previously for similar tasks (68% to 93%; see Rim et al., 2009, 2013; Todorov & Uleman, 2002). There were also noteworthy interindividual differences in accuracy (SD = 13%), which could be scored somewhat reliably (corrected split-half reliability: r = .61, 95% CI [0.53, 0.67]).

Tracing Positivity Differences

Our first research question was whether positivity differences would manifest in our recognition task at all. Basically, our hope was to put perceivers in a situation where they would likely make many spontaneous trait inferences such that we could examine if some perceivers preferentially inferred desirable traits, whereas others preferentially inferred undesirable ones. As intended, participants indeed made many spontaneous trait inferences. As can be seen in the left part of Figure 3, even though there were exactly six undesirable and six desirable adjectives that had actually been mentioned in combination with the suggested target, most participants made many more recognitions. On average, they recognized roughly twice as many undesirable adjectives and roughly three times as many desirable adjectives as were actually presented to them in combination with the respective target in the study phase. There were also quite large interindividual differences in the number of recognized

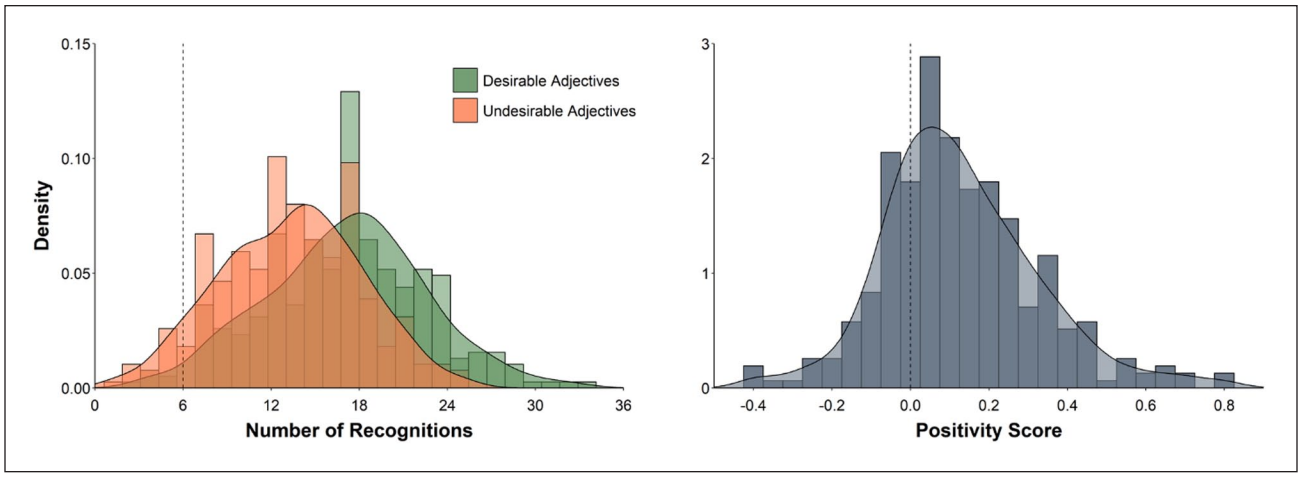

Figure 3. Distribution of recognitions for trials involving desirable vs. undesirable adjectives (left panel) and positivity scores (right panel) in Study 1. The dashed vertical lines represent the expected number of recognitions (left panel) and the expected positivity score (right panel) for a perfectly accurate perceiver.

adjectives, such that the distributions in the left part of Figure 3 span almost the entire range of possible recognitions. Most importantly, when we computed each participant's positivity score by subtracting the number of recognitions for undesirable adjectives from the number of recognitions for desirable adjectives and dividing this by their total number of recognitions, these scores also covered a quite large range of values, specifically from −0.40 to 0.80 ($M = 0.13$, $SD = 0.20$; see right part of Figure 3). Thus, even though the vignettes had been balanced in terms of valence, some participants had preferentially encoded negative personality aspects about the targets into their memory, whereas others had preferentially encoded positive personality aspects about them.

To estimate the precision of the positivity score, we computed its split-half reliability. For this, we divided the 72 recognition trials into parallel halves, each consisting of the same number of correctly versus incorrectly paired, explicit versus implicit, and desirable versus undesirable trials. Correlating the positivity scores from each half and correcting for the reduced length of the task, we found a moderate reliability estimate of $r = .58$, 95% CI [0.50, 0.65].

Correlates of Positivity Differences

To explore the psychological meaning of the positivity score, we computed Pearson correlations with participants' recognition accuracy, their explicit liking toward targets, and with the assessed personality constructs and demographic variables. Additionally, we computed Pearson correlations corrected for attenuation to account for the limited reliability. The corrected correlations can be interpreted as estimations of the associations if the positivity score had been measured with perfect reliability.

Results are shown in Table 1. First, there was a noteworthy negative association between the positivity score and recognition accuracy, such that perceivers who were better at making correct recognitions tended to recognize more negative traits compared to less accurate perceivers. This correlation was quite sizable once disattenuated, given that the reliabilities of both involved variables were rather modest. Concerning explicit liking, as could be expected, participants with more positive recognitions also reported liking the targets more. Concerning personality traits, individuals higher in agreeableness and conscientiousness were found to have higher positivity scores. These small to moderate associations were more pronounced once corrected for attenuation. Associations with extraversion, emotional stability, openness, and narcissistic admiration were not statistically significant. Other than expected, there was also no significant negative association with narcissistic rivalry. Concerning sociodemographic variables, we found a link between the positivity score and age: The older the participants were, the more positive their average recognitions. Finally, neither gender nor education was associated with the positivity score.

Study 2

The results of Study 1 were promising in that people differed systematically in the positivity of personality content they encoded when forming first impressions of unfamiliar targets under conditions that did not require explicit evaluative judgments. However, Study 1 also had several limitations, which we sought to address in Study 2.

First, because the reliability of the test scores was only moderate, we developed a more extensive and psychometrically improved version of the recognition task and examined whether the initial findings would replicate in a larger and more diverse sample.

Second, Study 1 could not rule out the possibility that positivity differences partly reflected processes during retrieval rather than earlier stages of impression formation.

Table 1. Nomological Network of Positivity Differences Measured in the Recognition Task.

Variable	*r* (*r'*)	*p*	95% CI
Recognition accuracy	**−.36** (−.61)	<.001	[−0.45, −0.26]
Liking towards targets	**.21** (.27)	<.001	[0.10, 0.31]
Personality traits			
Extraversion	.03 (.05)	.542	[−0.08, 0.15]
Agreeableness	**.14** (.24)	.012	[0.03, 0.25]
Conscientiousness	**.20** (.30)	<.001	[0.09, 0.31]
Emotional stability	.08 (.12)	.137	[−0.03, 0.19]
Openness	.09 (.15)	.124	[−0.02, 0.20]
Narcissistic admiration	.03 (.04)	.605	[−0.08, 0.14]
Narcissistic rivalry	−.04 (−.05)	.470	[−0.15, 0.07]
Sociodemographic variables			
Age	**.24** (.31)	<.001	[0.13, 0.34]
Gender[a] (0 = male, 1 = female)	.07 (.09)	.210	[−0.04, 0.18]
Education[b]	−.06 (−.08)	.165	[−0.17, 0.05]

Note. *r* = zero-order Pearson correlation. Coefficients in bold are significantly different from zero ($p < .05$). Correlations corrected for attenuation are presented in parentheses. The correction accounted for the low reliability of the positivity score. For correlations of the positivity score with the Big Five personality traits and recognition accuracy, the correction additionally accounted for measurement error in the criterion variable.
95% CI = 95% confidence intervals [lower limit, upper limit].
[a]Point biserial correlation coefficients were computed because the gender variable was dichotomous. The third gender category (i.e., diverse) had only one participant, who was excluded here.
[b]Kendall's correlation coefficients were computed because the education variable was ordinal.

In particular, participants may have differed in how they responded under uncertainty during the test phase. For example, some perceivers might have been more inclined to endorse positive traits and reject negative traits when unsure. To address this possibility, half of the participants in Study 2 were provided with an additional response option ("don't know") in the test phase, allowing us to examine whether positivity differences persisted when strategic guessing under uncertainty was reduced.

Third, we improved the basis for evaluating the nomological network of the recognition-based positivity measure. Specifically, we administered a well-validated Big Five questionnaire instead of the ad hoc measure used in Study 1, and additionally assessed self-esteem and attachment styles to examine associations with broader indicators of psychological adjustment.

Fourth, Study 1 did not directly compare the recognition-based measure with established rating-based approaches to assessing perceiver positivity. To examine the degree of overlap between positivity differences observed under lower evaluative task demands and those observed in explicit evaluative judgments, Study 2 additionally included a self-report measure of attitudes toward other people in general as well as a classic rating-based task of perceiver positivity. Finally, Study 1 used a convenience sample in which female, young, and highly educated participants were overrepresented. Study 2 used a much more heterogeneous and balanced sample in terms of gender, age, and education to improve on the generalizability of findings. Study 2 was not preregistered.

Method

Recognition Task

Stimuli. To extend the initial pool of stimuli, we generated and pretested 38 additional trait-implying sentences with 43 German speakers (19 male, 23 female, 1 diverse; M_{age} [*SD*] = 29.84 [8.28] years) who indicated up to three synonymous traits they would attribute to the target person presented in the sentence. Sentences that yielded high consensus regarding the implied traits were kept as they were and used for implicit trials (27 sentences; $M_{consensus}$ [*SD*] = 64.16 [18.54]%), and the remaining sentences were amended with the respective adjective and used for explicit trials. Photographs, names, and ages of targets were created in the same fashion as for the original pool of targets. In sum, this yielded 19 additional vignettes.

Test Phase. Whereas the study phase was identical to that in Study 1 (apart from covering more targets), a few slight changes were made to the test phase. Since this phase consisted of more than twice as many trials as in Study 1 (148 trials) and would have required the participants to click through overly many survey pages, all four trials pertaining to a given target were presented at once, rather than one by one. Further, participants were randomly assigned to two conditions with different response options for answering which of the adjectives had occurred in the vignettes: In the first condition there were two response options (1 = *occurred*; 2 = *did not occur*), in the second condition there were three response options (1 = *occurred*; 2 = *did not occur*; 3 = *I don't know*).

Participants. The data collection was carried out in cooperation with a professional panel provider, and the sample size was determined by available recruitment resources. With the planned budget, we recruited the largest sample feasible and evaluated what effect size this affords to detect with conventional power. Our final sample (n = 831) yields 80% power (two-sided α = .05) to detect correlations of approximately $|r| \approx .11$. Participants were compensated with 5€ for completing the entire study. Overall, 1,322 participants completed the study and confirmed that they were native German speakers and between 18 and 70 years old. Since each participant was assigned an individual ID by the panel provider, it was possible to identify and exclude 60 participants whose ID was recorded multiple times. Further, 39 participants who took less than 10 min to complete the entire study and 386 participants who showed conspicuous response behavior were excluded. We deemed response patterns conspicuous if more than 85% of the trials had been answered with the same response (i.e., classifying adjectives as [not] having occurred during the study phase) or if more than 95% of the trials had been answered with "I don't know." The remaining sample consisted of 831 participants (462 identifying as male, 368 as female, 1 as diverse) who ranged from age 18 to 70 (M = 46.1, SD = 14.6) and came from heterogenous educational backgrounds as reported on a 6-point scale (1 = *no graduation* [0%]; 2 = *graduated after 8th grade* [2%]; 3 = *graduated after 10th grade* [17%]; 4 = *graduated from senior high school* [15%]; 5 = *apprenticeship* [31%]; 6 = *bachelor´s degree or higher* [32%]). Again, participants' race or ethnicity was not assessed.

Procedures. At the beginning of the online survey, participants were informed that the study was an investigation of the relationship between personality and music preference. After providing sociodemographic information, participants completed the recognition task described above. They then completed all of the validation measures listed below except for the rating-based positivity measure. Finally, participants were informed of the actual purpose of the study. To assess rating-based positivity, all participants who completed the main survey were re-invited approximately 1 week later to participate in a follow-up survey containing the Online-Tool for Assessing Perceiver Effects (O-TAPE; Rau, Nestler, et al., 2021). The reason why the O-TAPE was assessed in a separate survey was to avoid response fatigue.

Validation Measures

Explicit Liking. On the last two pages of the survey, participants were presented with the targets and asked "How likeable was this person to you?" using a drop-down menu showing a 6-point Likert-style scale ranging from 1 (*very unlikeable*) to 6 (*very likeable*). As before, the mean score in liking across targets was computed for each participant (α = .90).

Big Five. The Big Five personality traits were assessed with the German version of the Big Five Inventory-2-Short Form (BFI-2-S; Rammstedt et al., 2020). The BFI-2-S consists of 30 Items and covers the Big Five personality traits Extraversion (e.g., "I am someone who tends to be quiet."), Agreeableness (e.g., "I am someone who is compassionate, has a soft heart."), Conscientiousness (e.g., "I am someone who tends to be disorganized."), Negative Emotionality (formerly Neuroticism; e.g., "I am someone who worries a lot.") and Open-Mindedness (formerly Openness to Experience; e.g., "I am someone who is fascinated by art, music, or literature.") with six items per domain. Responses were coded on a 5-point Likert-type scale ranging from 1 (*disagree strongly*) to 5 (*agree strongly*; α_E = .71; α_A = .63; α_C = .78; α_{ES} = .81; α_O = .71).

Narcissism. Narcissism was again assessed with the NARQ-S (Back et al., 2013; $\alpha_{admiration}$ = .76; $\alpha_{rivalry}$ = .61).

Self-Esteem. Self-Esteem was assessed with the Single-Item Self-Esteem Scale (Robins et al., 2001). Participants rated "I have high self-esteem." on an 8-point Likert-style scale ranging from 1 (*does not apply at all*) to 8 (*applies perfectly;* estimated reliability = .75; see Heise, 1969).

Attachment Style. Attachment Styles were assessed with a German translation of Bartholomew and Horowitz's (1991) measure, as in Rau, Nestler, et al. (2021). The scale contains brief descriptions of four attachment prototypes: *fearful, dismissing, preoccupied, and secure* (e.g., for secure attachment: "It is easy for me to become emotionally close to others. I am comfortable depending on others and having others depend on me. I don't worry about being alone or having others not accept me."). The agreement of the participants to each statement was measured on a 6-point Likert-type scale ranging from 1 (*does not apply at all*) to 6 (*applies perfectly*). The *secure* prototype describes security in close emotional relationships and dependencies. The *dismissing* prototype describes the refusal of close emotional relationships due to a preference for independence and autonomy. The *preoccupied* prototype describes the conflicting desire for close emotional relationships but fear of rejection. The *anxious* prototype describes an avoidance of intimate relationships due to a lack of trust in others.

Explicit Anthropologic Beliefs. Participants' explicit views of human nature were measured with an ultra-short adaption of the *Philosophies of Human Nature* questionnaire (Wrightsman, 1964) as in Rau, Nestler, et al. (2021). The measure consists of six items capturing people's assumptions about human nature regarding *trustworthiness, altruism, independence, strength of will, complexity*, and *diversity*. Each item consists of description of the underlying dimension's high and low poles (e.g., for altruism: "In some situations, humans act completely selflessly and are

solely concerned with benefitting others" and "Eventually, humans always act in their own interest, and they are never exclusively concerned with benefitting others") and assesses a rating from 1 (*perfect agreement with the statement on the left*) to 6 (*perfect agreement with the statement on the right*). Since *complexity* and *diversity* are not indications of a person's general positive or negative attitude (Wrightsman, 1964), we followed the approach of Rau, Nestler, et al. (2021) and averaged only the remaining four items to index global positivity of explicit anthropologic beliefs ($\alpha = .58$).

Rating-Based Positivity. To assess the perceiver positivity as measured in a classic rating-based manner, we used the O-TAPE (Rau, Nestler, et al., 2021). The O-TAPE is an online tool presenting 10 screenshots of Facebook profiles of different targets, balanced in terms of gender, age, ethnicity, attractiveness, and expressiveness. To rate the targets, a six-item bipolar adjective scale developed by Thielmann, et al. (2023) was used. It contains six pairs of adjectives, representing the high and low poles of the HEXACO dimensions respectively (see Lee & Ashton, 2004, 2008): *greedy, dishonest* versus *modest, sincere* (Honesty-Humility), *anxious, sensitive* versus *relaxed, emotionally stable* (Emotional Stability), *reserved, quiet* versus *sociable, lively* (Extraversion), *critical, quick-tempered* versus *sympathetic, gentle* (Agreeableness), *disorganized, careless* versus *reliable, self- disciplined* (Conscientiousness), and *conventional, uncreative* versus *open to new experiences, artistic* (Openness to Experience). To control for acquiescence variance (i.e., perceivers preferring positive response options on a scale) we reversed the anchors for three of the six scales and later recoded them (cf. Thielmann et al., 2023). The follow-up survey was completed by 381 participants, 40 of whom were excluded due to conspicuous response behavior. The remaining 341 participants ranged from age 19 to age 70 ($M = 48.9$, $SD = 13.9$) and consisted of 214 individuals identifying as male (63%), 126 as females (37%), and one as diverse (<1%). Their educational background was similar to the full sample (1 = *no graduation* [0%]; 2 = *graduated after 8th grade* [6%]; 3 = *graduated after 10th grade* [13%]; 4 = *graduated from senior high school* [13%]; 5 = *apprenticeship* [34%]; 6 = *bachelor´s degree or higher* [34%]). The O-TAPE positivity scores were highly reliable, $\alpha = .84$).

Results and Discussion

Preliminary Analysis

As in Study 1, a higher recognition rate for correctly paired (vs. falsely paired) trials was found both in trials in which the trait had actually been presented and in trials in which the trait had only been implied. As shown in Figure 2, this applies both for participants who, as in Study 1, were forced to guess in case of uncertainty and to participants who were given a "don't know" response option.

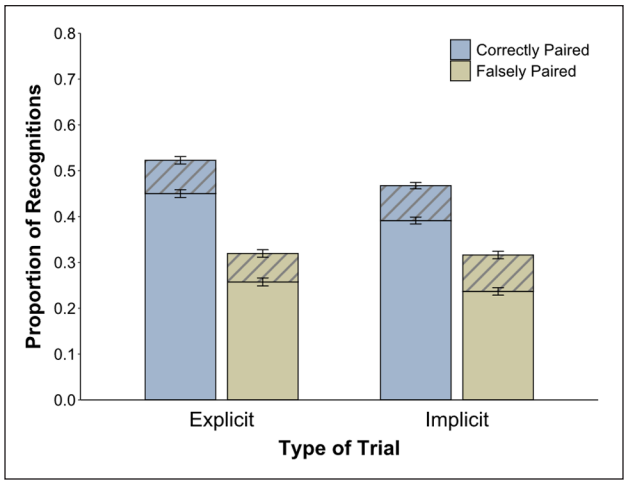

Figure 4. Mean recognition rates (with standard errors) by type of trial in Study 2. Hatched areas show the difference in mean recognition rates between participants with vs. without a "don't know" response option. Trials in which adjectives had actually been mentioned are shown on the left side. Trials in which adjectives had only been implied are shown on the left side.

Explicit Trials. Overall, the mean proportion of correct recognitions of explicitly presented traits in filler trials ($M = 0.60$; $SD = 0.11$) significantly exceeded the chance level of 0.50, $t(830) = 25.99$, $p < .001$, $d = 0.90$. This applied to participants who were forced to guess under uncertainty, $M = 60$; $SD = 0.11$, $t(460) = 19.18$, $p < .001$, $d = 0.89$, and to those who were given a "don't know" option, $M = 0.60$, $SD = 0.11$, $t(375) = 17.57$, $p < .001$, $d = 0.91$. Also, there was no significant difference in the mean proportion of correct recognitions of explicitly presented traits in filler trials between these two groups, $t(812) = 0.69$, $p = .245$, $d = 0.05$. Of note, the recognition accuracy of 60% is somewhat lower than in Study 1 and in similar tasks used previously, which is likely a consequence of the increased length and, hence, increased difficulty of the task.

Implicit Trials. As in Study 1, participants were more likely to falsely recognize implied traits that were paired correctly with the target ($M = 0.43$, $SD = 0.15$) than falsely paired with a different, random target ($M = 0.28$, $SD = 0.17$), $t(830) = 29.19$, $p < .001$, $d = 0.94$. Again, this applied both to participants who were forced to guess in case of uncertainty, $t(460) = 20.79$, $p < .001$, $d = 0.93$, and to participants who were offered a "don't know" option in the test phase, $t(369) = 20.63$, $p < .001$, $d = 1.03$.

In sum, as can be seen from inspecting the hatched areas in Figure 4, the "don't know" option was not used excessively overall and was used similarly across the different types of trials. As such, whether or not participants were forced to guess in cases of uncertainty did not affect the fundamental functioning of the task, which is more consistent with the interpretation that variance in recognition responses

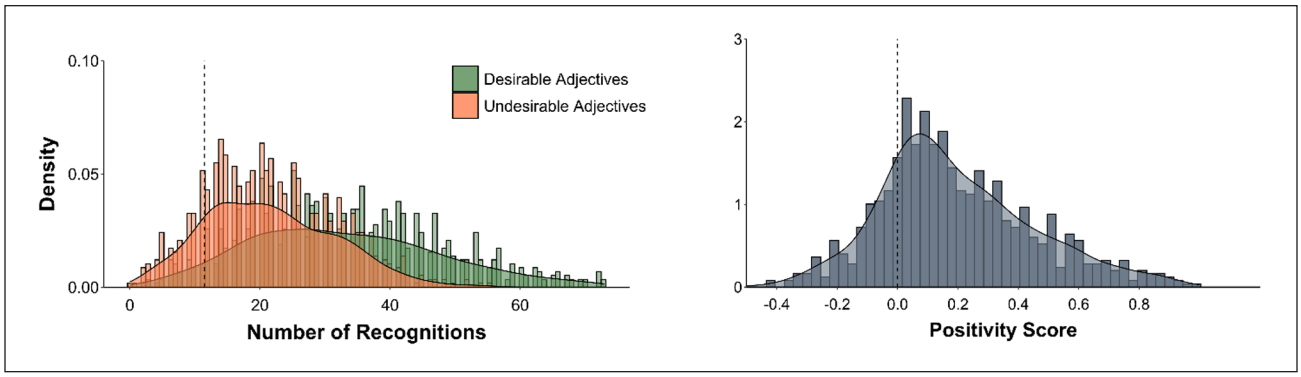

Figure 5. Distribution of recognitions for trials involving desirable vs. undesirable adjectives (left panel) and positivity scores (right panel) in Study 2. The dashed vertical lines represent the expected number of recognitions (left panel) and the expected positivity score (right panel) for a perfectly accurate perceiver.

reflects differences arising during earlier stages of impression formation rather than strategic differences at retrieval.

Tracing Positivity Differences

Figure 5 shows the distribution of recognitions for trials with desirable vs. undesirable traits and the distribution of positivity scores. As in Study 1, participants differed greatly in terms of the positivity of recognized traits. Importantly, positivity scores of participants who were forced to guess under uncertainty ($M = 0.19$, $SD = 0.26$) were essentially identical to those of participants who could respond "don't know" ($M = 0.19$, $SD = 0.26$). Specifically, neither the means nor the variances differed significantly between these groups, $t(788) = 0.33$, $p = .371$, $d = 0.02$; $F(460, 369) = 0.98$, $p = .867$. This pattern indicates that positivity differences were not driven by differential positive versus negative guessing during the test phase, and is consistent with the interpretation that they reflect differences arising during earlier stages of impression formation.

As in Study 1, we determined the reliability of the positivity score by correlating two parallel test halves. The Spearman-Brown corrected split-half estimate suggested that reliability was high, $r = 0.81$, 95% CI [0.78, 0.83].

Correlates of Positivity Differences

Table 2 provides a within-sample comparison of the nomological networks of positivity assessed via the recognition task versus deliberate trait ratings (O-TAPE). Here, nomological consistency refers to the similarity of these two correlational profiles across the same set of external criteria (i.e., the correlation between the vectors of criterion correlations; Thielmann & Hilbig, 2019). In the present data, the two profiles showed moderate nomological consistency ($r = .63$), suggesting substantial overlap in what the two operationalizations capture, while also leaving room for systematic divergences. Beyond the similarity in shape (i.e., which correlates are relatively stronger vs. weaker), Table 2 also suggests a modest difference in the overall magnitude of criterion correlations. Averaging across the 18 shared criteria using Fisher r–to–z transformation (and back-transformation), the mean absolute association was $|r| = .13$ for recognition-task positivity and $|r| = .16$ for O-TAPE positivity. Thus, O-TAPE positivity tended to show somewhat stronger criterion associations overall, although the difference was small in absolute terms.

At the level of specific correlates, most associations were directionally aligned and similar in size, including links with anthropological beliefs, most Big Five traits, and several attachment dimensions. However, six correlates differed significantly between the two positivity measures. First, recognition-task positivity was strongly negatively related to recognition accuracy ($r = -.35$), whereas O-TAPE positivity was essentially unrelated to accuracy ($r = .02$), $\Delta r = -.37$. Second, O-TAPE positivity showed stronger positive associations with liking of the targets ($r = .43$ vs. .31) and agreeableness ($r = .37$ vs. .14). Third, O-TAPE positivity showed more negative associations with narcissistic rivalry ($r = -.30$ vs. $-.12$) and with secure attachment ($r = -.19$ vs. .04). Finally, O-TAPE positivity was higher among women ($r = .23$) whereas recognition-task positivity showed essentially no association with gender ($r = -.01$). Overall, recognition-based positivity was uniquely intertwined with recognition accuracy, whereas rating-based (O-TAPE) positivity showed more consistent links with interpersonal/evaluative dispositions (e.g., liking, agreeableness). However, because the two measures were obtained in different stimulus contexts and from different targets, these contrasts should be interpreted as within-sample differences between recognition-task and deliberate-rating operationalizations of positivity, not as a clean decomposition of processing stages.

Supplementary Analysis: Heterogeneity in Positivity Across Trial Types

To explore whether perceivers differ in trial-type–specific "positivity styles" (e.g., positivity expressed primarily for

Table 2. Nomological Network of Positivity Differences Measured in the Recognition Task and in Personality Ratings of Strangers (O-TAPE).

Criterion variable	Recognition task positivity score (n = 831)			O-TAPE positivity score (n = 341)			Difference in correlations		
	r	p	95% CI	r	p	95% CI	Δr	p	95% CI
Recognition accuracy	**−.35**	<.001	[−0.41, −0.29]	.02	.715	[−0.09, 0.13]	**−.37**	<.001	[−0.49, −0.25]
Liking towards targets	**.31**	<.001	[0.25, 0.37]	**.43**	<.001	[0.34, 0.51]	**−.12**	.026	[−0.23, −0.01]
Positivity of anthropologic beliefs	**.17**	<.001	[0.10, 0.23]	**.24**	<.001	[0.13, 0.33]	−.07	.268	[−0.19,0.05]
Personality traits									
Extraversion	**.17**	<.001	[0.10, 0.23]	.10	.070	[−0.01, 0.20]	.07	.278	[−0.06, 0.19]
Agreeableness	**.14**	<.001	[0.07, 0.21]	**.37**	<.001	[0.27, 0.45]	**−.22**	<.001	[−0.34, −0.11]
Conscientiousness	**.17**	<.001	[0.10, 0.23]	**.23**	<.001	[0.12, 0.33]	−.06	.338	[−0.18; 0.06]
Emotional stability	**.16**	<.001	[0.09, 0.22]	.10	.066	[−0.01, 0.20]	.06	.384	[−0.07, 0.18]
Openness	.03	.399	[−0.04, 0.10]	.06	.271	[−0.05, 0.16]	−.03	.637	[−0.16, 0.10]
Narcissistic admiration	.06	.096	[−0.01, 0.13]	−.04	.410	[−0.15, 0.06]	.10	.112	[−0.02, 0.23]
Narcissistic rivalry	**−.12**	.001	[−0.19, −0.05	**−.30**	<.001	[−0.39, −0.20]	**.18**	.004	[0.06, 0.29]
Psychological adjustment									
Self-esteem	**.10**	.003	[0.04, 0.17]	.09	.095	[−0.02, 0.19]	.01	.835	[−0.11, 0.14]
Secure attachment	.04	.313	[−0.03, 0.10]	**−.19**	<.001	[−0.29, −0.09]	**.23**	<.001	[0.10, 0.35]
Preoccupied attachment	.02	.478	[−0.04, 0.09]	−.02	.709	[−0.13, 0.09]	.04	.487	[−0.08, 0.17]
Dismissing attachment	**−.11**	.002	[−0.17, −0.04]	−.11	.038	[−0.22, −0.01]	.01	.923	[−0.12, 0.13]
Fearful attachment	**.17**	<.001	[0.10, 0.24]	.09	.098	[−0.02, 0.19]	.08	.196	[−0.04, 0.21]
Sociodemographic variables									
Age	**.15**	<.001	[0.08, 0.22]	**.17**	.001	[0.07, 0.27]	−.02	.720	[−0.14, 0.10]
Gender[a] (0 = male, 1 = female)	−.01	.720	[−0.08, 0.06]	**.23**	<.001	[0.13, 0.33]	**−.24**	<.001	[−0.36, −0.12]
Education[b]	**−.06**	.013	[−0.11, −0.01]	−.11	.080	[−0.18, −0.03]	.04	.350	[−0.04, 0.13]

Note. r = zero-order Pearson correlation. 95% CI = 95% confidence intervals [lower limit, upper limit]. Coefficients in bold are significantly different from zero ($p < .05$).
[a]one diverse-gender participant was excluded in this analysis. The third gender category (i.e., diverse) had only one participant, who was excluded here.
[b]Kendall's correlation coefficients were computed because the education variable was ordinal.
The correlation between both positivity scores is r = .19 [0.09, 0.29], $p < .001$. Because O-TAPE was assessed in a follow-up completed by a subset of participants, we report a same-sample version of this table (recomputing recognition-task correlations in the O-TAPE subsample) in the online supplement; results are unchanged, suggesting the comparison is not driven by attrition.
O-TAPE = Online-Tool for Assessing Perceiver Effects.

explicit vs. implicit trials or for correctly vs. incorrectly paired trials), we conducted additional multilevel logistic regression analyses on the trial-level data. Comparing random-effects structures indicated that the best-fitting model included a participant-specific random slope for trait valence (positive vs. negative), suggesting trial-general individual differences in positivity. Models that additionally allowed positivity to vary by explicitness or pairing accuracy (i.e., random slopes for positive × explicit and/or positive × correct) did not improve fit. Full model specifications and comparisons are reported in the online supplement.

General Discussion

This research examined whether perceiver positivity in first impressions is already evident during early stages of impression formation, under conditions that do not require explicit evaluative judgment. Results from two studies using a newly developed recognition task revealed that even when perceivers are not explicitly asked to indicate their impressions of others, they vary substantially in the desirability of the trait content they encode about them. Thus, even though the task minimized concerns related to politeness, benevolence, or self-presentational concerns in explicit judgments, we observed considerable interindividual differences in the positivity of first impressions, suggesting that such differences can emerge during early stages of impression formation. Study 2 replicated and extended these findings using a more reliable task, a more diverse sample, and a richer set of validation measures. The inclusion of a "don't know" response option further indicated that positivity differences were not driven by guessing behavior, and is consistent with an account that emphasizes contributions of early-stage processing.

Previous research on perceiver effects has often assumed that positivity differences emerge early in impression

formation, before explicit evaluative judgments are required (Kenny, 2019; Rau, Carlson, et al., 2022; Rau et al., 2023; Srivastava et al., 2010; Wood et al., 2010), but a direct empirical test of this assumption had been missing. Results from Study 2 revealed a noticeable but moderate association between positivity as typically captured in the perceiver effects literature (i.e., via personality ratings) and positivity as measured with the recognition-based task. This pattern suggests that early-stage processes contribute to perceiver positivity, but also that rating-based measures reflect additional variance beyond what is captured under lower evaluative task demands. Accordingly, distinguishing between positivity that emerges during early impression formation and positivity expressed in explicit evaluative judgments may help refine theories of impression formation and perceiver effects. This distinction aligns with stage-based models of social cognition, which differentiate early encoding of person information from later evaluative and judgmental processes (e.g., Fiske & Neuberg, 1990).

A closer comparison of recognition-based and rating-based positivity further clarifies which correlates of perceiver positivity appear to reflect early-stage impression formation and which seem to emerge primarily at later, evaluative stages. Several of the most established correlates of rating-based positivity—namely agreeableness, narcissistic rivalry, and explicit liking of targets (Rau et al., 2021; Heynicke et al., 2021)—re-emerged in the present data and were evident for both operationalizations. However, these associations were consistently weaker for recognition-based than for rating-based positivity. This attenuation suggests that these constructs are not purely rooted in early perceptual encoding but likely exert a substantial portion of their influence at later stages, when impressions are translated into deliberate evaluations. For instance, agreeable individuals may indeed encode somewhat more positive information about others, but their pronounced positivity in rating-based tasks likely also reflects norms of benevolence, interpersonal motivation, or self-presentational concerns. Similarly, narcissistic rivalry may be only weakly linked to early-stage negativity, whereas its stronger association with explicit ratings plausibly reflects strategic devaluation of others at the judgment stage. In this sense, recognition-based positivity appears to capture a more basic perceptual tendency that is subsequently amplified, modulated, or overridden by motivational and normative factors captured by rating-based measures.

A particularly clear dissociation between early- and later-stage positivity emerged for gender. Replicating prior work, women showed higher positivity than men in rating-based judgments (Rau et al., 2021; Winquist et al., 1998), but no corresponding difference was observed for recognition-based positivity. This pattern is consistent with the interpretation that the well-documented "female positivity" effect reflects gender-stereotypical behavior in explicit social judgment—such as greater communal orientation, politeness norms, or concern with social harmony—rather than differences in spontaneous encoding of person information. In other words, women may not form more positive initial impressions than men, but may be more likely to express positivity when asked to evaluate others explicitly.

Finally, secure attachment showed a negative association with rating-based positivity but not with recognition-based positivity. This finding is difficult to interpret, not least because prior studies using rating-based measures have not consistently reported negative links between secure attachment and positivity (Rau, Carlson, et al., 2022; Srivastava et al., 2010). Given this lack of convergence, the present result should be treated with caution and may reflect sample-specific artifact rather than a theoretically meaningful dissociation between early- and later-stage positivity.

Future Directions

Our recognition task yielded robust evidence that perceiver positivity is already evident during early stages of impression formation under conditions of reduced evaluative task demands and, in its longer form used in Study 2, the task was reliable enough to warrant future research on how early-stage positivity relates to other individual differences relevant to impression formation and interpersonal relations. Specifically, future work should explore whether the task has predictive validity for real-life interpersonal dynamics observed in naturalistic study settings such as round-robin designs. For instance, early-stage positivity may predict interpersonal approach behavior (e.g., smiling, active listening) during the getting-acquainted phase of dyads or groups, perhaps even incrementally above positivity expressed in explicit evaluative judgments. This possibility should be explored in future studies. At the same time, although the present findings are consistent with an early-stage account of perceiver positivity, the recognition paradigm does not allow a strict separation between differences arising during encoding versus retrieval, a distinction that future research could address through additional constraints during the encoding phase (e.g., time pressure or concurrent cognitive load).

An important limitation of the novel recognition task is that, although vignettes were balanced on valence (one desirable and one undesirable behavior per target), positive and negative behaviors were not strictly matched on other dimensions that can shape dispositional inference and memory (e.g., perceived diagnosticity, traitedness, and linguistic abstraction; Skowronski & Carlston, 1989; Semin & Fiedler, 1988). Such asymmetries may influence mean recognition rates and the overall positivity offset. However, because the present research focuses on individual differences within a fixed stimulus set, our central conclusions do not depend on interpreting the task in absolute terms.

Finally, further research is warranted to examine the long-term consequences that early-stage positivity may have for social relationships. According to Contemporary Integrative

Interpersonal Theory (Hopwood, 2018; Hopwood et al., 2021; Pincus & Ansell, 2013), recurring interpersonal dynamics are a key aspect of both normal personality and personality pathology, and these dynamics are thought to be fueled by people's perceptions of others (Rau et al., 2023). As such, our recognition task may offer a tool for identifying a potential source of the characteristic "interpersonal transaction cycles" that different people tend to experience.

Constraints on Generality. All participants were German speakers, and the targets used as stimuli were representative of the population of German adults. As such, generalizations across sociodemographic groups and particularly to non-WEIRD societies should be made with caution.

Conclusion

Across two studies, this work investigated the roots of positivity differences in person perception. To this end, we developed a novel recognition task that minimizes explicit evaluative judgment demands. Results revealed considerable interindividual variability in the positivity of personality impressions formed under these conditions, providing initial evidence that perceiver positivity is not solely a function of intentional judgment but is already evident during early stages of impression formation. This distinction may inform theories of impression formation and may also be of clinical relevance for identifying and addressing perception biases in psychopathology. Future research that further refines the methodological approach and more precisely characterizes the respective contributions of early-stage processing and explicit evaluative judgments is warranted.

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Ethical Considerations

All study procedures were approved by the ethics committee of the University of Münster (protocol number 2021-25-RR). All participants provided informed consent prior to their participation.

Consent for Publication

The individuals depicted in Figure 1 provided informed consent for the publication of their images.

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Declaration of Conflicting Interests

The authors declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this article

Data Availability Statement

The data and analysis code to reproduce all of the results reported in this article are retrievable from https://osf.io/64389/.

Supplemental Material

Supplemental material is available at https://osf.io/64389/.

Notes

1. Our goal was to minimize semantic proximity among trials. For instance, we sought to avoid to first ask whether a certain target had been described as *lenient* (an indicator of high agreeableness) and then ask whether the same target had been described as *resentful* (an indicator for low agreeableness) since recognizing the former would probably reduce the chance of recognizing the latter. To verify that sufficient independence was achieved, we pretested all 108 adjective pairs to the 33 pretest participants and asked them to indicate semantic similarity on a 5-point scale (−2 [*very opposite*]; 0 [*independent*] +2 [*very similar*]). As intended, the pairs were rated as largely independent (M = 0.05, SD = 0.39).
2. We also explored an alternative positivity score which accounted for different degrees of valence in traits. Specifically, instead of *counting* the number of recognitions, we weighted each recognition by the respective trait's valence rating retrieved from the Aachen List of Trait Words (Britz et al., 2019). The resulting score was highly similar to the unweighted positivity score (r = .99, 95% CI [0.99, 0.99]) and yielded essentially identical results concerning our research questions. For simplicity, we focus on the unweighted score in the main text and present results based on the weighted positivity score online (OS 3).

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